Yuvraj Dutta

Curriculum Vitae

EDUCATION

Master of Statistics [M.Stat] (Probability Specialization) **2025–Present**
Indian Statistical Institute, Kolkata First-Year Score: **92.2%**
· *Dissertation: Asymptotic Fluctuations in p-spin Ising Models* Rank: 3/67
Advisors: Prof. Sumit Mukherjee (Columbia University) and Prof. Arijit Chakrabarti (ISI Kolkata)

Bachelor of Statistics [B.Stat] (Hons.) **2022–2025**
Indian Statistical Institute, Kolkata Aggregate Score: **90.3%**
· *Awarded the **D. Basu Memorial Gold Medal** as the Most Outstanding Student in B.Stat Programme.* Rank: 4/41

RESEARCH INTERESTS

My research interest lies in developing mathematical frameworks motivated by applied sciences, with particular interest in statistical physics, stochastic processes and high-dimensional statistics.

MANUSCRIPTS & PREPRINTS

[1] Dutta, Y., and Mukherjee, S. (2026). *Fluctuations in p-spin Ising Models* URL: `Manuscript in Preparation`.

[2] Aurzada, F., Dutta, Y., and Wiegand, M. (2026). *Brownian Motion with Drift conditioned to have Restricted L^2-Norm.* Submitted to the *Applied Probability journals.* URL: `Preprint`.

[3] Eich, Y., Dutta, Y., and Koeppl, H. (2026). *Entropic Matching for Filtering and Smoothing of Stochastic Differential Equations.* Submitted to a conference. URL: `Manuscript available upon request`.

[4] Dutta, Y., Barui, S., Mitra, D., and Balakrishnan, N. (2026). *Bivariate Inverse Gaussian Degradation Processes with Shared Random Effects and an Application to Fatigue Cracks.* Submitted to *Applied Stochastic Models in Business and Industry.* URL: `Preprint`.

RESEARCH EXPERIENCE

Fluctuations in p-spin Ising Models (Master's Dissertation) *Ongoing*
Supervisor: Prof. Sumit Mukherjee *Columbia University, New York*

· Studied fluctuations of global magnetisation in p-spin Ising models in the high-temperature regime and proved Berry-Esseen bounds using Stein's method of exchangeable pairs.

· These models are particularly important since they capture higher-order network motifs, generalizing frameworks like the edge-triangle ERGM.

Brownian Motion with Drift conditioned to have Restricted L^2-Norm `[Preprint]` *Feb 2026 – Sep 2026*
Supervisor: Prof. Frank Aurzada *Technische Universität, Darmstadt*

· Studied Brownian motion with T-dependent drift conditioned on an atypically small L^2-norm, proving convergence of the conditioned process to an Ornstein-Uhlenbeck process independent of the original drift.

· Derived asymptotics for the small ball probability of the quadratic additive functional via the Bromwich integral and saddle-point techniques. Identified the limiting conditioned process using Girsanov's theorem.

Entropic Matching for Filtering and Smoothing of Stochastic Differential Equations *Feb 2025 – Dec 2025*
Supervisor: Prof. Heinz Koeppl *Technische Universität, Darmstadt*

· Extended entropic matching to approximate filtering and smoothing distributions for latent continuous-time stochastic differential equation models with discrete observations, adapting the variational projection framework to non-Gaussian posterior families.

· Derived tractable filtering and smoothing dynamics for lognormal and Gamma approximations, and incorporated expectation propagation for non-conjugate observations.

Bivariate Inverse Gaussian Degradation Processes with Shared Random Effects `[Preprint]` *May 2024 – Oct 2024*
Supervisor: Prof. Sandip Barui *ISI, Kolkata*

· Introduced a bivariate degradation model using inverse Gaussian processes with a generalized gamma shared frailty and developed likelihood and Monte Carlo-based estimation procedures. Derived an approximation for system reliability.

· Evaluated their performance through simulation and applied the model to fatigue-crack data.

Limiting Free Entropy Density of the Mean-Field Ashkin-Teller Model `[Report]` *Dec 2025 - Feb 2026*
Supervisors: Profs. Somabha Mukherjee and Bhaswar B. Bhattacharya *NUS/University of Pennsylvania*

· Derived the limiting free entropy density using the method of types and entropy bounds, following the classical Curie-Weiss approach, and expressed the limit as a finite dimensional optimisation problem.

INVITED TALKS AND SEMINARS

- Tracking the Hidden States: Bayesian Filtering & Smoothing [Abstract]/ [Slides] *Oct 2025*
  D. Basu Memorial Gold Medal Award Seminar, ISI Kolkata
- Brownian Motion with Drift Conditioned to Have Restricted L^2-Norm *Jun 2026*
  Departmental Research Seminar, Department of Mathematics, TU Darmstadt
- Entropic Matching for Filtering and Smoothing of Stochastic Differential Equations *Jul 2025*
  Self-Organizing Systems Lab Seminar, TU Darmstadt

ACADEMIC HONORS

- Awarded the **D. Basu Memorial Gold Medal** , presented to the **Most Outstanding Student** in the B.Stat (Hons.) Programme 2022–25 at the Indian Statistical Institute, Kolkata. *2025*
- Received the **Scholarship for Academic Excellence** in 2nd-5th semesters of the B.Stat programme and all semesters of the M.Stat programme. *2022–26*

RESEARCH INTERNSHIPS

- Selected for an on-site Summer Research Internship at the *Stochastics Group, Department of Mathematics, Technische Universität Darmstadt, Germany*, under Prof. Frank Aurzada. *2026*
- Selected for an on-site Summer Research Internship at the *Self-Organizing Systems Lab, Technische Universität Darmstadt, Germany*, under Prof. Heinz Koeppl. *2025*

SELECTED PROJECTS

- Reading Project on Theory of Large Deviations — Prof. Arijit Chakrabarty, ISI Kolkata *May - Jul 2023*
- Reading Project on e-Values and Multiple Testing — Prof. Bowen Gang, Fudan University *Jan - Apr 2024*
- Statistical Depth and Robust Multivariate Inference [Report] — Multivariate Analysis Course *Nov 2025*
- Resampling-Based Inference for Log-Linear Models [Report] — Resampling Techniques Course *Mar 2026*
- Bootstrap-Based Model Selection in Binary GLMs [Report] — Resampling Techniques Course *Apr 2026*
- Factorial Design for Gameplay Performance [Report] — Design of Experiments Course *May 2026*

ACHIEVEMENTS

- Ranked **8th nationally** and **24th overall** in the Pairs Category (with Nishant Lamboria) of the **Simon Marais Mathematics Competition** (East Division), an international undergraduate mathematics competition. *2024*
- Achieved **Regional Rank 11** in the Madhava Mathematics Competition, a nationwide proof-based mathematics competition for undergraduate students. *2023*
- Qualified for the Indian National Mathematical Olympiad **(INMO) 2021**, precursor to IMO, after ranking among the top 9 students regionally. Qualified for RMO 2018-20. *2018–21*
- Secured **AIR 50** in the ISI B.Stat Entrance Examination, **AIR 1322** in JEE Main, and **AIR 1557** in JEE Advanced among 1 million+ candidates nationwide. *2022*
- Awarded the Kishore Vaigyanik Protsahan Yojana **(KVPY) Fellowship**, a national scholarship programme for students demonstrating exceptional aptitude for scientific research. *2021*
- Secured **98.0%** in the **CBSE Class X** Board Exam and **97.8%** in the **CBSE Class XII** Board Exam. *2020, 2022*

STANDARDIZED TEST SCORES

TOEFL iBT **118/120**
GRE **326/340** (Quantitative: **170/170**, Verbal: **156/170**)

SKILLS

Programming Python, R, C
Scientific Computing NumPy, SciPy, PyTorch
Typesetting LaTeX

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